Single-Normal empirical-Bayes shrinkage vs. ashr

Is this shrinkage the same as `ashr`?

The candidate formula:

$$\sigma^2_{\text{prior}} \;=\; \max\!\Big(\,\overline{\hat\beta^2} \;-\; \overline{se^2},\; 0\Big)$$

$$\hat\beta_{\text{shrunk}} \;=\; \hat\beta \cdot \frac{\sigma^2_{\text{prior}}}{\sigma^2_{\text{prior}} + se^2}$$

Short answer: same family, but `ashr` is strictly more general.

What the formula above is

This is single-Normal empirical Bayes (a James–Stein / `limma`-style shrinker):

- Prior: $\beta \sim \mathcal{N}(0,\, \sigma^2_{\text{prior}})$
- Likelihood: $\hat\beta \mid \beta \sim \mathcal{N}(\beta,\, se^2)$
- Posterior mean: $\hat\beta \cdot \sigma^2_{\text{prior}}/(\sigma^2_{\text{prior}} + se^2)$
- σ^2_{prior} estimated by method of moments: $\mathrm{Var}(\hat\beta) \approx \sigma^2_{\text{prior}} + \overline{se^2}$, so $\sigma^2_{\text{prior}} \approx \overline{\hat\beta^2} - \overline{se^2}$, clipped at 0.

One scalar prior variance, one Normal component, no null spike.

How `ashr` (Stephens 2017) differs

1. **Mixture prior, not a single Normal.** `ashr` fits

   $$g(\beta) \;=\; \pi_0\,\delta_0 \;+\; \sum_k \pi_k\,\mathcal{N}(0,\, \sigma_k^2)$$

   (or uniform components) — a flexible **unimodal** prior. The $\{\pi_k, \sigma_k\}$ are estimated by EM. The effective shrinkage factor is therefore not $\sigma^2/(\sigma^2 + se^2)$ everywhere; it is a mixture-weighted blend that depends on where $\hat\beta$ falls relative to each component.

2. **Explicit point mass at 0** ($\pi_0\,\delta_0$). This is what lets `ashr` produce **lfsr** (local false sign rate) and **lfdr** (local false discovery rate). The single-Normal formula above has no null component — it shrinks everything toward 0 by a smooth factor but never assigns "probably exactly zero" mass.

3. **Different shrinkage shape.** With a single Normal prior, the shrinkage factor depends only on se^2 (the same σ^2_{prior} for every observation). `ashr`'s posterior mean is more aggressive near zero (the null component pulls hard) and gentler in the tails (heavier-tailed components dominate). Two observations with the same se but different $|\hat{\beta}|$ are **differently** shrunk by `ashr`; under the single-Normal formula they receive the same proportional shrinkage.

When the two coincide

The single-Normal formula = `ashr` restricted to one mixture component ($K = 1$), $\pi_0 = 0$, and σ_1 fixed by method-of-moments instead of MLE. In that degenerate case the two posterior-mean formulas agree exactly.

Practical implication for this project

If `PosteriorMeanMatrices/` was produced by R's `ashr` (via `SRC/RScripts/run_ashr_on_chunk.R`), the values will differ from what the single-Normal formula would give — especially for:

- genes near zero (`ashr` shrinks **more**, because the π_0 component pulls hard),
- large-effect genes (`ashr` shrinks **less**, because heavier-tailed mixture components dominate).

The two will agree in spirit (rank order roughly preserved), not in numbers.